

Search For The **ORGANIC BLUE** Is In Full Swing



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Benefits of organic light emitting diodes (OLEDs) go beyond simple static image quality and the responsiveness and smoothness of the display itself. These are capable of refresh rates a 1000 times faster than standard LED-backlit liquid crystal display (LCD) panels. So these can easily catch a blur that might have been lost in traditional displays. But are we happy with such blazing-fast displays?

No, we definitely are not. Numerous researches are underway to further improve OLED efficiency and illumination at a lower cost. After all, a reduced price point sells. It helps boost displays for televisions, phones, tablets and more to an entire new level.

Like LCDs, OLEDs produce colour through green, red and blue sub-pixels on screen. However, there is a catch.

Blue emitters are hard to find

Organic molecules emitting blue light have been rare to find. Dr Mark Thompson, who is Ray R. Irani chair of chemistry, department of chemistry, chemical engineering and materials science, University of South California, USA, explains, "Blue phosphorescence remains a central challenge and has been so for the last ten years."

Manufacturers have created organometallic molecules to overcome this problem.

The problem here lies with expensive transition metals like iridium. These are used to enhance the molecule through phosphorescence, in order to improve efficiency. This solution, though expensive, does not produce stable blue colour.

Advanced matrix OLED (AMOLED) and super AMOLED has been one of the interesting application areas. Some of the best displays in the recent past have been the super-AMOLED ones like that of Samsung Galaxy Note 7.

Towards the new blue

Molecular Space Shuttle, a large-scale computer-driven screening process, was developed by a research team from Harvard University, USA, in collaboration with Massachusetts Institute of Technology, and Samsung. It identifies new OLED molecules that perform at par with industry standards through machine learning, cheminformatics, theoretical and experimental chemistry.

Kyulux, an advanced materials startup working in the field of OLED displays and lighting technology, has also announced securing Molecular Space Shuttle deep-learning system license for the discovery of materials for display and lighting applications. However, this process is more suited for finding dopants and not base materials.

An interdisciplinary team led by Dr Alan Aspuru-Guzik, professor of chemistry and chemical biology, Harvard University, has been working on replacing these organometallic systems with entirely organic molecules. Some algorithms were developed to compute the likelihood of efficient outcomes on a candidate pool of about 1.6 million. This helped prioritise molecules to be virtually tested and significantly reduce computational costs. Hopefully, these could help bring the prices down for hi-end devices like a MacBook Pro or Galaxy.

Flexibility in OLEDs

